

Chapter 2: Literature Review / Project Evaluation

2.1 Feedback & Evaluation

The Student Performance Dashboard project demonstrates a practical application of programming and database concepts in the educational domain. The integration of Python for data preprocessing and SQL for structured data storage reflects a well-aligned approach with modern data pipeline practices.

Strengths of the project include:

- Relevance of the problem: Monitoring student performance is a real-world challenge.
- Technical integration: Combines Python and SQL effectively.
- Structured workflow: Clear milestones for development.
- Data handling: Proper preprocessing techniques such as cleaning and categorization.

Areas for improvement include:

- Limited validation of data accuracy
- Basic visualization level
- Lack of advanced analytics

2.2 Suggested Improvements

1. Data Quality Enhancement:

- Implement validation rules
- Handle missing data using advanced techniques

2. Advanced Data Analysis:

- Add statistical analysis
- Apply basic machine learning models

3. Improved Database Design:

- Normalize tables
- Add constraints (primary/foreign keys)

4. Enhanced Visualization:

- Use Plotly or advanced charts
- Add filters to dashboard

5. User Experience:

- Add dropdowns and KPIs

6. Documentation & Presentation:

- Include screenshots
- Explain code and SQL queries clearly

2.3 Final Grading Criteria

Documentation – 25%

Implementation – 30%

Data Processing – 15%

SQL & Database Design – 10%

Visualization – 10%

Testing & Validation – 5%

Presentation – 5%

Total – 100%

2.4 Summary

This chapter evaluated the Student Performance Dashboard project. While it demonstrates strong foundational skills, adding advanced analytics, better visualization, and improved user interaction can enhance its effectiveness.